51. A rope by which a calf is tied is increased from 12 m to 23 m. How much additional grassy ground shall it graze?
(a) 1120 m^2 (b) 1250 m^2
(c) 1210 m^2 (d) 1200 m^2

52. Four circular cardboard pieces, each of radius 7 cm are placed in such a way that each piece touches two other pieces. The area of the space enclosed by the four pieces is:
(a) 21 cm^2 (b) 42 cm^2
(c) 84 cm^2 (d) 168 cm^2

53. The length and breadth of a rectangular field are in the ratio 5:3. If the cost of cultivating the field at 25 paise per square metre is ₹6000, then find the dimensions of the field:
(a) 250 m, 100 m (b) 50 m 30 m
(c) 200 m, 120 m (d) Cannot be determined

54. The cost of carpeting a room 5 m wide with carpet at ₹3.50 per m^2 is ₹105. The length of the room is:
(a) 3.5 m (b) 5 m
(c) 6 m (d) 6.5 m

55. The length of a rectangular field is twice its breadth. If the rent of the field at ₹3500 per hectare is ₹28000, then find the cost of surrounding it with fence at ₹5 per metre.
(a) ₹6000 (b) ₹7000
(c) ₹6500 (d) ₹8000

56. The area of a rectangular field is 27000 m^2. The ratio of its length and breadth is 6:5. The length and breadth of the field are respectively:
(a) 180 m, 150 m (b) 200 m, 150 m
(c) 180 m, 120 m (d) 150 m, 100 m

57. The area of a sector of a circle of radius 5 cm, formed by an arc of length 3.5 cms, is:
(a) 35 cm^2 (b) 17.5 cm^2
(c) 8.75 cm^2 (d) 55 cm^2

58. The length of a plot is double its width. If a square piece of land of area 150 m^2 occupies $\frac{1}{3}$ area of the plot, then what is the length of the plot?
(a) 15 m (b) 7.5 m
(c) 30 m (d) 10 m

59. A wire is in the form of a semi-circle of 7 cm radius. The length of the wire will be:
(a) 25 cm (b) 36 cm
(c) 5 cm (d) 69 cm

60. A circular road runs round a circular ground. If the difference between the circumferences of the outer circle and the inner circle is 66 m, the width of the road is:
(a) 21 m (b) 10.5 m
(c) 7 m (d) 5.25 m

61. If the area of a square is 50 sq units, then the area of the circle drawn on its diagonal is:
(a) 25 π sq units (b) 50 π sq units
(c) 100 π sq units (d) None of these

62. A rectangular sheet of cardboard is of 4 cm × 2 cm. If a circle of greatest possible area is cut from it, then the area of remaining portion is:
(a) $(2 - \pi)$ cm^2 (b) $(4 - \pi)$ cm^2
(c) $(8 - \pi)$ cm^2 (d) $(16 - \pi)$ cm^2

63. What is the radius of a circle, to the nearest cm, whose area is equal to the sum of the areas of the three circles of radii 22 cm, 19 cm and 8 cm, respectively?
(a) 17 cm (b) 30 cm
(c) 29 cm (d) 19 cm

64. The length of a rectangle is increased by 33.33%. By what per cent should the width be decreased to maintain the same area?
(a) 25% (b) 33.33%
(c) 22.5% (d) None of these

65. If the area of a square is equal to the area of a rectangle 6.4 m long and 2.5 m wide, then each side of the square measures:
(a) 8 m (b) 5.4 m
(c) 3.8 m (d) 4 m

66. The diameters of two concentric circles are 8 cm and 10 cm. The area of the region between them is:
(a) π cm^2 (b) 3π cm^2
(c) 6π cm^2 (d) 9π cm^2

67. The length of a rectangular room is 4 m. If it can be partitioned into two equal square rooms, what is the length of the partition in metres?
(a) 1 (b) 2
(c) 4 (d) Data inadequate

68. A rectangular carpet has an area of 120 m^2 and a perimeter of 46 m. The length of its diagonal is:
(a) 15 m (b) 16 m
(c) 17 m (d) 20 m

69. The side of a square is 22 m. What is the radius of the circle whose circumference is equal to the perimeter of the square?

(a) 28 m (b) 3.5 m
(c) 14 m (d) 7 m

70. A piece of wire 132 cm long is bent successively in the shapes of an equilateral triangle, a square, a regular hexagon, and a circle. Then, which has the largest surface area?
(a) Equilateral triangle
(b) Square
(c) Circle
(d) Regular hexagon

71. If the radius of one circle is twelve times the radius of another, how many times does the area of the greater contain the area of the smaller?
(a) 12 (b) 72
(c) 144 (d) 96

72. If the circumference of a circle is equal to the perimeter of a square, what is the ratio of the area of the circle to the area of the square?
(a) 22:7 (b) 14:11
(c) 11:7 (d) 4:1

73. The length of a rectangular plot of land is three times as much as its breadth. A playground measuring 1200 ft^2 occupies $\frac{1}{4}$ of the total area of the plot. What is the length of the plot in feet?
(a) 40 (b) 360
(c) 120 (d) Data inadequate

74. There are two squares s_1 and s_2. The ratio of their areas is 4:25. If the side of s_1 is 6 cm, what is the side of s_2?
(a) 20 cm (b) 15 cm
(c) 5 cm (d) 12 cm

75. The radius of the wheel of a vehicle is 70 cm. The wheel makes 10 revolutions in 5 seconds. The speed of the vehicle is:
(a) 29.46 Km/h (b) 31.68 Km/h
(c) 36.25 Km/h (d) 32.72 Km/h

76. A rectangular carpet has an area of 60 m^2. Its diagonal and longer side together equal 5 times the shorter side. The length of the carpet is:
(a) 5 m (b) 13 m
(c) 14.5 m (d) 12 m

77. A playground has the shape of a rectangle with two semi-circles on its smaller sides as diameters, added outside. If the sides of the rectangle are 36 m and 24.5 m, then the area of the playground is: $\left(\text{use } \pi = \frac{22}{7}\right)$
(a) 2259.529 m^2 (b) 1353.625 m^2
(c) 1139.523 m^2 (d) None of these

78. A man runs around a circle of 50 m radius at a speed of 12 Km/h. Find the time taken by him for going around it ten times:
(a) 10 minutes (b) 12.5 minutes
(c) 15.7 minutes (d) None of these

79. A room 5 m × 8 m is to be carpeted leaving a margin of 10 cm from each wall. If the cost of the carpet is ₹18 per m^2, then the cost of carpeting the room will be:
(a) ₹702.60 (b) ₹691.80
(c) ₹682.46 (d) ₹673.92

80. The area of a big rectangle is equal to the area of a small rectangle. If the length of the big rectangle is equal to the length of the small rectangle and the width of big rectangle is 2 m, what is the width of a small rectangle?
(a) $\frac{1}{3}$ m (b) 1 m
(c) 2 m (d) Cannot be determined
(e) None of these

81. If the radius of a circle is reduced by 40%, then its circumference is reduced by:
(a) 60% (b) 40%
(c) 35% (d) 45%

82. A figure consists of a square of side 'a' m with semi-circles drawn on the outside of the square. The area (in m^2) of the figure so formed will be:
(a) a^2 (b) $a^2 + 2\pi a^2$
(c) $4\pi a^2$ (d) $a^2 + \frac{\pi a^2}{2}$

83. The area of a square field is 6050 m^2. How much time it will take to reach from one of its corner to the opposite corner at the rate of 10 m in every 30 seconds?
(a) $5\frac{1}{2}$ minutes (b) 11 minutes
(c) 22 minutes (d) 110 minutes

84. If a regular hexagon is inscribed in a circle of radius r, then its perimeter is:
(a) $3r$ (b) $6r$
(c) $9r$ (d) $12r$

85. 2 poles 15 m and 30 m high stand upright in a playground. If their feet be 36 m apart, find the distance between their tops.

(a) 36 m (b) 39 m
(c) 15 m (d) None of these

86. The length of a rectangular hall is $\frac{4}{3}$ of its width. If the area of the hall is 300 m^2, what is the difference between the length and the breadth?
(a) 15 m (b) 20 m
(c) 3 m (d) 5 m
(e) 6 m

87. Each side of an equilateral triangle is increased by 1.5%. The percentage increase in its area is:
(a) 1.5% (b) 3%
(c) 4.5% (d) 5.7%

88. A rope, by which a horse is tied, is increased from 12 to 23 m. How much additional ground will it be able to graze?
(a) 1315 m^2 (b) 765 m^2
(c) 1210 m^2 (d) 1012 m^2

89. If a diagonal of a square is doubled, how does the area of the square change?
(a) Becomes 4-fold (b) Becomes 3-fold
(c) Becomes 2-fold (d) None of these

90. If the sides of a rectangle are increased by 20%, the percentage increase in its perimeter is:
(a) 80 (b) 40
(c) 20 (d) None of these

91. A circle road runs around a circular garden. If the difference between the circumference of the outer circle and the inner circle is 44 m, then find the width of the road.
(a) 4 m (b) 7 m
(c) 3.5 m (d) 7.5 m

92. The length of a rectangular field is double its width. Inside the field there is a square-shaped pond 8 m long. If the area of the pond is $\frac{1}{8}$ of the area of the field, what is the length of the field?
(a) 32 m (b) 64 m
(c) 16 m (d) 20 m
(e) None of these

93. A circular disc of area 0.49π m^2 rolls down a length of 1.76 Km. The number of revolutions it makes is:
(a) 300 (b) 400
(c) 600 (d) 4000

94. If area of a triangle whose base is 6 cm is equal to the area of a circle of radius 6 cm, then find the height of this triangle:
(a) 10 cm (b) 22 cm
(c) 12 cm (d) 18 cm

95. The length of a ladder exactly equals the height of a wall. If the ladder is placed on a 2-feet tall stool placed 10 feet away from the wall, its tip just touches the top of the wall. The height of the wall in feet is:
(a) 15 (b) 26
(c) 28 (d) 32

96. If the diagonal of a square is doubled to make another square, the area of the new square will:
(a) Become 4-fold (b) Become 2-fold
(c) Become 6-fold (d) Become 8-fold

97. The area of a circle is 154 cm^2. The length of an arc of the circle which subtends an angle of 45° at the centre is:
(a) 11 cm (b) 5.5 cm
(c) 7 cm (d) None of these

98. A lawn is in the form of a triangle having its base and height in the ratio 2:3. The area of the lawn is $\frac{1}{12}$ hectare. Find the base and height of the lawn.
(a) 55 m, 34 m (b) 50 m, $33\frac{1}{3}$ m
(c) 50 m, 35 m (d) Data inadequate

99. A rectangular farm has to be fenced on one long side, one short side and the diagonal. If the cost of fencing is ₹10 per m, the area of the farm is 1200 m^2 and the short side is 30 m long, how much would the job cost?
(a) ₹700 (b) ₹1200
(c) ₹1400 (d) ₹1500

100. The diameter of a circle is 105 cm less than the circumference. What is the diameter of circle?
(a) 44 cm (b) 46 cm
(c) 48 cm (d) 49 cm

101. A garden is 24 m long and 14 m wide. There is a path 1 m wide outside the garden along its sides. If the path is to be constructed with square marble tiles 20 cm × 20 cm, then find the number of tiles required to cover the path:
(a) 1800 (b) 200
(c) 2000 (d) 2150

102. If the length of the diagonal of a rhombus is 80% of the length of the other diagonal, the area of the rhombus is how many times the square of the length of the longer diagonal?

(a) $\frac{4}{5}$ (b) $\frac{2}{5}$

(c) $\frac{3}{4}$ (d) $\frac{1}{4}$

103. *ABCD* is a trapezium in which $AB \parallel CD$ and $AB = 2CD$. If its diagonals intersect each other at *O*, the ratio of the area of triangle *AOB* and *COD* is:

(a) 1:2 (b) 2:1
(c) 1:4 (d) 4:1

104. The ratio of the corresponding sides of two similar triangles is 3:4. The ratio of their areas is:

(a) 4:3 (b) 3:4
(c) 9:16 (d) $\sqrt{3}:2$

105. The area of the circle inscribed in an equilateral triangle of side 24 cm is:

(a) 24 π cm^2 (b) 36 π cm^2
(c) 48 π cm^2 (d) 18 π cm^2

106. The radius of wheel is 1.4 decimetre. How many times does it revolve during a journey of 0.66 Km?

(a) 375 (b) 750
(c) 1500 (d) 3000

EXERCISE-2
(BASED ON MEMORY)

1. If the length of one side and the diagonal of a rectangle are 7 cm and 25 cm respectively, then find its perimeter (in cm).

(a) 124 (b) 36
(c) 62 (d) 72

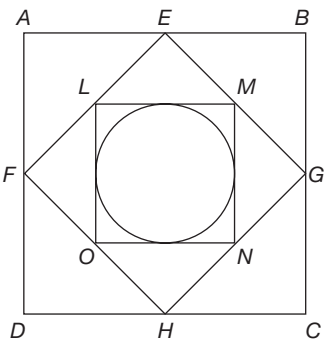
[SSC CHSL (10+2) Tier-I CBE, 2018]

2. Find the perimeter (in cm) of a semicircle od radius 28 cm.

(a) 288 (b) 144
(c) 121 (d) 242

[SSC CHSL (10+2) Tier-I CBE, 2018]

3. In the given figure. ABCD is a square. EFGH is a square formed by joining mid points of sides of ABCD. LMNO is a square fanned by joining mid points of sides of EFGH. A circle is. Inscribed inside EFGH. If area of circles is 38.5 cm^2 then what is the area (in cm^2) of square ABCD?



(a) 98 (b) 196
(c) 122.5 (d) 171.5

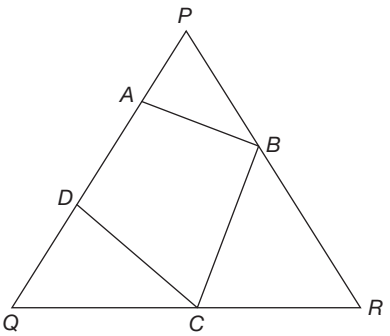
[SSC CGL Tier-II CBE, 2018]

4. ABC is a triangle. AB = 5 cm. AC = $\sqrt{41}$ cm. and BC = 8 cm. AD is perpendicular to BC. What is the area (in cm^2) of triangle ABD?

(a) 12 (b) 6
(c) 10 (d) 20

[SSC CGL Tier-II CBE, 2018]

5. In the given figure, PQR is a triangle and quadrilateral ABCD is inscribed in it QD = 2 cm. QC = 5cm. CR = 3 cm. BR = 4 cm. PB = 6cm. What is the area (In cm^2) of the quadrilateral ABCD?



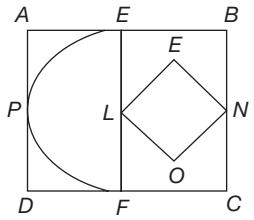
(a) $\frac{(23\sqrt{21})}{4}$ (b) $\frac{(15\sqrt{21})}{4}$

(c) $\frac{(17\sqrt{21})}{5}$ (d) $\frac{(23\sqrt{21})}{5}$

[SSC CGL Tier-II CBE, 2018]

6. In the given figure, ABCD is a square of side 14 cm. E and F are mid-points of sides AB and DC

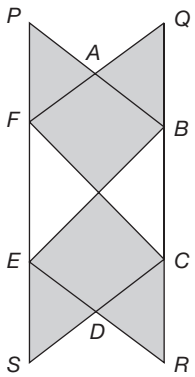
respectively. EPF is a semicircle whose diameter is EF. LMNO is square. What is the area (In cm^2) of the shaded region?



(a) 108.5 (b) 94.5
(c) 70 (d) 120

[SSC CGL Tier-II CBE, 2018]

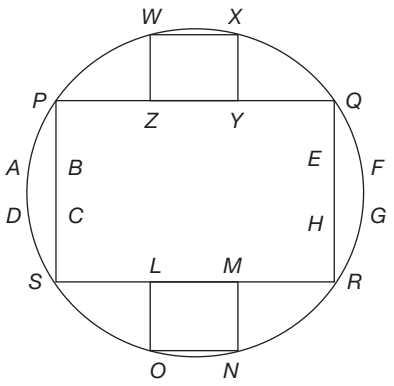
7. In the given figure, ABCDEF is a regular hexagon whose side is 6 cm. APF, QAB, DCR and DES are equilateral triangles. What is the area (In cm^2.) of the shaded region?



(a) $24\sqrt{3}$ (b) $18\sqrt{3}$
(c) $72\sqrt{3}$ (d) $36\sqrt{3}$

[SSC CGL Tier-II CBE, 2018]

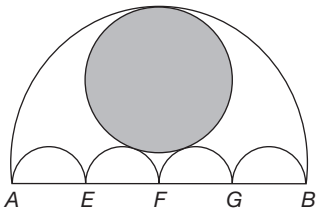
8. In the given figure radius of a circle is $14\sqrt{2}$ cm. PQRS is a square. *EFGH*, *ABCD*. *WXYZ* and *LMNO* are four identical squares. What Is the total area (In cm^2) of all the small squares?



(a) 31.36 (b) 125.44
(c) 62.72 (d) 156.8

[SSC CGL Tier-II CBE, 2018]

9. In the given figure, *AB AE*. *EF*, *FG* and *GB* are semicircles. *AB* = 56 cm. and *AE* = *EF* = *FG* = *GB*. What is the area (In cm) of the shaded region?



(a) 414.46 (b) 382.82
(c) 406.48 (d) 394.24

[SSC CGL Tier-II CBE, 2018]

10. The length and breadth of a rectangle are increased by 10% and 20% respectively. What will be the percentage increase in the area of rectangle?

(a) 30% (b) 32%
(c) 28% (d) 33%

[SSC CHSL (10+2) Tier-I CBE, 2018]

11. A circular wire of length 168 cm is cut and bent in the form of a rectangle whose sides are in the ratio of 5:7. What is the length (in cm) of the diagonal of the rectangle?

(a) $\sqrt{4127}$ (b) $\sqrt{3137}$

(c) $\sqrt{1813}$ (d) $\sqrt{3626}$

[SSC CAPFs ASI & Delhi Police SI, 2017]

12. ΔABC is similar to ΔPQR. Length of AB is 16 cm and length of the corresponding side PQ is 9 cm. If area of ΔABC is 1024sq cm, what is the area of ΔPQR?

(a) 768 sq. cm. (b) 32 sq. cm
(c) 324 sq. cm. (d) 128 sq. cm

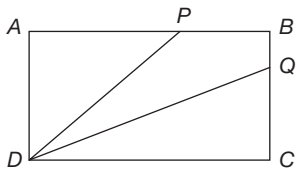
[SSC CHSL (10+2) Tier-I CBE, 2017]

13. In ΔABC, a line parallel to side BC cuts the sides *AB* and *AC* at points *D* and E respectively and also point *D* divides *AB* In the ratio of 1:4. If area of ΔABC is 200 cm^2, then what is the area (in cm^2) of quadrilateral *DECB*?

(a) 192 (b) 50
(c) 120 (d) 96

[SSC CAPFs ASI & Delhi Police SI, 2017]

14. In the given figure, PB is one-third of AB and BQ is one-third of BC. If the area of BPDQ is 20 cm^2 then what is the area (in cm^2) of ABCD?



(a) 45 (b) 30
(c) 40 (d) 60

[SSC CAPFs ASI & Delhi Police SI, 2017]

15. The area of a circle is same as the area of a square. What is the ratio of the diameter of the circle and diagonal of the square?

(a) $1:\sqrt{\pi}$ (b) $2:\sqrt{\pi}$
(c) $\sqrt{2}:\sqrt{\pi}$ (d) 1: π

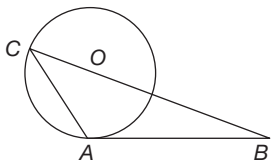
[SSC CAPFs ASI & Delhi Police SI, 2017]

16. In ΔABC, AD and AE are bisectors of $\angle BAC$ and $\angle BAD$ respectively. If $\angle BAE = 30°$, $AE = 9$ cm and $EC = 15$ cm, what is the area (In cm^2) of ΔAEC?

(a) 36 (b) 54
(c) 72 (d) 216

[SSC CAPFs ASI & Delhi Police SI, 2017]

17. In the given figure, triangle ABC is drawn such that AB is tangent to a circle at A whose radius is 10 cm and BC passes through centre of the circle. Point C lies on the .circle. If BC = 36 cm and AB = 24 cm, then what is the area (in cm^2) of triangle ABC?



(a) 134.5 (b) 148
(c) 168 (d) 180

[SSC CGL Tier-I CBE, 2017]

18. The tangents drawn at points A and B of a circle with centre O, meet at P If ∠AOB = 120° .and AP = 6 cm, then what is the area of triangle APB (in cm^2)?

(a) $6\sqrt{3}$ (b) $8\sqrt{3}$
(c) 9 (d) $9\sqrt{3}$

[SSC CGL Tier-I CBE, 2017]

19. D and E are points on sides and AC of ΔABC. DE is parallel to BC. If.AD: DB = 2: 3, what is the ratio of area of ΔADE and area of quadrilateral BDEC?

(a) 4: 21 (b) 4: 25
(c) 4: 29 (d) 4: 9

[SSC CHSL (10+2) Tier-I CBE, 2017]

20. Three circles of radius 21 cm are placed in such a way· that each circle touches the other two. What is the area of the portion enclosed by the three circles? (In sq. cm)

(a) $441\sqrt{3}-693$ (b) $882\sqrt{3}-693$
(c) $882\sqrt{3}-462$ (d) $441\sqrt{3}-462$

[SSC CAPFs ASI & Delhi Police SI, 2017]

21. If the area of a square is increased by 44%, retaining its shape as a square, each of its sides increases by:

(a) 19% (b) 21%
(c) 22% (d) 20%

[SSC CAPFs (CPO) SI & ASI Delhi Police, 2017]

22. A cylindrical vessel of height 5 cm and radius 4 cm is completely filled with sand. When this sand is poured out it forms a right circular cone of radius 6 cm. what will be the height of this cone?

(Take $\pi + \frac{22}{7}$)

(a) 6.67 cm (b) 2.22 cm
(c) 3.33 cm (d) 1.67 cm

[SSC CHSL (10+2) Tier-I (CBE), 2017]

23. A solid metallic sphere of radius 21 cm is melted and recast into a cone with diameter of the base as 21 cm. What is the height (in cm) of the cone?

(a) 336 (b) 112
(c) 224 (d) 66

[SSC CAPFs ASI & Delhi Police SI, 2017]

24. Three equal circles of unit radius touch one another. Then the area of the circle circumscribing in the three circles is

(a) $6\pi(2+\sqrt{3})^2$ (b) $\frac{\pi}{6}(2+\sqrt{3})^2$
(c) $\frac{\pi}{3}(2+\sqrt{3})^2$ (d) $3\pi(2+\sqrt{3})^2$

[SSC CPO, 2016]

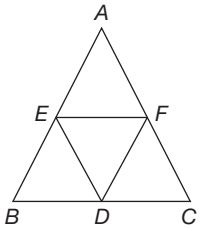
25. In Δ ABC and Δ PQR ∠B = ∠Q, ∠C = ∠R. M is the mid-point of side QR. If AB:PQ = 7:4, then $\frac{\text{area}\,(\Delta ABC)}{\text{area}(\Delta PMR)}$ is

(a) $\frac{1}{2}$ (b) 1

(c) $\frac{1}{3}$ (d) $\frac{1}{4}$

[SSC CAPFs (CPO) SI & ASI, Delhi Police, 2016]

26. If D, E and F are the mid-points of the sides of an equilateral triangle ABC, then the ratio of the area of triangle DEF and DCF is:

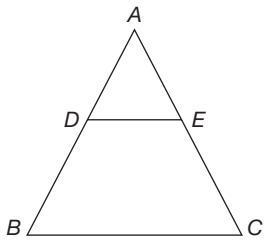


(a) 1.1:1 (b) 1:1.1

(c) 0.9:1 (d) 1:1

[SSC CGL Tier-I (CBE), 2016]

27. In figure, $DE \parallel BC$. If DE = 3cm BC = 6 cm and area of ΔADE = 15 sq. cm, then the area of ΔABC is



(a) 75 sq. cm. (b) 45 sq. cm.

(c) 30 sq. cm. (d) 60 sq. cm.

[SSC CGL Tier-I CBE, 2016]

28. The length and breadth of a rectangular place of a land are in a ratio 5: 3. The owner spent ₹6000 for surrounding it from all sides at ₹7.50 per metre. The difference between its length and breadth is

(a) 50 metre (b) 100 metre

(c) 150 metre (d) 250 metre

[SSC CGL Tier-II (CBE), 2016]

29. The length of the two parallel sides of a trapezium are 16m and 20m respectively. If its height is 10m its area in square metre is

(a) 360 (b) 260

(c) 240 (d) 180

[SSC CGL Tier-I (CBE), 2016]

30. Three medians AD, BE and CF of ΔABC intersect at G. The area of ΔABC is 36 sq. cm. Then the area of Δ CGE is

(a) 12 sq. cm. (b) 6 sq. cm.

(c) 9 sq. cm. (d) 18 sq. cm.

[SSC CGL Tier-I (CBE), 2016]

31. A rectangular plot has a concrete path running in the middle of the plot parallel to the breadth of the plot. The rest of the plot is used as a lawn, which has an area of 240 sq m. If the width of the path is 3m and the length of the path is greater than its breadth by 2m, what is the area of the rectangular plot? (in sq m)

(a) 255 (b) 168

(c) 288 (d) 360

(e) 224

[LIC, 2015]

32. The sides of a triangle having area 7776 sq. cm are in the ratio 3:4:5. The perimeter of the triangle is

(a) 400 cm (b) 412 cm

(c) 424 cm (d) 432 cm

[SSC, 2015]

33. In ΔABC, a line through A cuts the side BC at D such that BD:DC = 4:5. If the area of ΔABD = 60 cm^2, then the area of ΔADC is

(a) 50 cm^2 (b) 60 cm^2

(c) 75 cm^2 (d) 90 cm^2

[SSC, 2015]

34. The area of an isosceles trapezium is 176 cm^2 and the height is $2/11^{th}$ of the sum of its parallel sides. If the ratio of the length of the parallel sides is 4 : 7, then the length of a diagonal (in cm) is

(a) 28 (b) 24

(c) $\sqrt{137}$ (d) $2\sqrt{137}$

[SSC, 2015]

35. In trapezium ABCD, AB || CD and AB = 2CD. Its diagonals intersect at O. If the area of Δ AOB = 84 cm^2, then the area of Δ COD is equal to

(a) 21 cm^2 (b) 42 cm^2

(c) 72cm^2 (d) 26 cm^2

[SSC, 2015]

36. The perimeter of a rhombus is 60 cm and one of its diagonal is 24 cm. The area (in sq.cm) of the rhombus is

(a) 432 (b) 216

(c) 108 (d) 206

[SSC, 2015]

37. ABCD is a cyclic quadrilateral. AB and DC when produced meet at P, if PA = 8 cm, PB = 6 cm, PC = 4 cm, then the length (in cm) of PD is

(a) 8 (b) 10
(c) 12 (d) 6

[SSC, 2015]

38. Two circles touch externally. The sum of their areas is 130π sq cm and the distance between their centres is 14cm. The radius of the smaller circle is:

(a) 3 cm (b) 4 cm
(c) 5 cm (d) 2 cm

[SSC, 2015]

39. A circular swimming pool is surrounded by a concrete wall 4m wide. If the area of the concrete wall surrounding the pool is 11/25 that of the pool, then the radius (in m) of the pool is:

(a) 16 (b) 20
(c) 30 (d) 8

[SSC, 2015]

40. AB and CD are two parallel chords of a circle lying on the opposite side of the centre and the distance between them is 17cm. The length of AB and CD are 10cm and 24 cm respectively. The radius (in cm) of the circle is:-

(a) 18 (b) 15
(c) 13 (d) 9

[SSC, 2015]

41. A conical iron piece having diameter 28cm and height 30cm is totally immersed into the water of a cylindrical vessel, resulting in the rise of water level by 6.4cm. The diameter, in cm, of the vessel is:-

(a) 35 (b) 32
(c) 3.5 (d) $\frac{35}{2}$

[SSC, 2015]

42. If the altitude of an equilateral triangle is $12\sqrt{3}$ cm, then its area would be:

(a) 12 cm^2 (b) 72 cm^2
(c) $36\sqrt{3}$ cm^2 (d) $144\sqrt{3}$ cm^2

[SSC, 2015]

43. *ABCD* is a trapezium with *AD* and *BC* parallel sides. *E* is a point on *BC*. The ratio of the area of *ABCD* to that of *AED* is:

(a) $\frac{\overline{AD}}{\overline{BC}}$ (b) $\frac{\overline{BE}}{\overline{EC}}$
(c) $\frac{\overline{AD}+\overline{BE}}{\overline{AD}+\overline{CE}}$ (d) $\frac{\overline{AD}+\overline{BC}}{AD}$

[SSC, 2014]

44. In an equilateral triangle of side 24 cm, a circle is inscribed touching its sides. The area of the remaining portion of the triangle is $\left(\sqrt{3}=1.732\right)$:

(a) 98.55 cm^2 (b) 100 cm^2
(c) 101 cm^2 (d) 95 cm^2

[SSC, 2014]

45. Perimeter of a rhombus is 2*p* units and sum of length of diagonals is *m* units, then area of the rhombus is:

(a) $\frac{1}{4}m^2p$ sq units

(b) $\frac{1}{4}mp^2$ sq units

(c) $\frac{1}{4}(m^2-p^2)$ sq units

(d) $\frac{1}{4}(p^2-m^2)$ sq units

[SSC, 2014]

46. Two sides of a plot measuring 32 *m* and 24 *m* and the angle between them is a perfect right angle. The other two sides measure 25 *m* each and the other three angles are not right angles. The area of the plot in m^2 is:

(a) 768 (b) 534
(c) 696.5 (d) 684

[SSC, 2014]

47. A is the centre of circle whose radius is 8 and B is the centre of a circle whose diameter is 8. If these two circles touch externally, then the area of the circle with diameter AB is:

(a) 36 π (b) 64 π
(c) 144 π (d) 256 π

[SSC, 2014]

48. The perimeters of a circle, a square and an equilateral triangle are same and their areas are C, S and T respectively. Which of the following statement is true?

(a) C = S = T (b) C > S > T
(c) C < S < T (d) S < C < T

[SSC, 2014]

49. A wire of length 44 cm is first bent to form a circle and then rebent to form a square. The difference of the two enclosed areas is

(a) 44 cm^2 (b) 33 cm^2
(c) 55 cm^2 (d) 66 cm^2

[SSC, 2014]

50. ABCD is a parallelogram in which diagonals AC and BD intersect at O. If E, F, G and H are the mid points of AO, DO, CO and BO respectively, then the ratio of the perimeter of the quadrilateral EFGH to the perimeter of parallelogram ABCD is

(a) 1 : 4 (b) 2 : 3
(c) 1 : 2 (d) 1 : 3

[SSC, 2014]

51. On decreasing each side of an equilateral triangle by 2 cm, there is a decrease of $4\sqrt{3}$ cm^2 in its area. The length of each side of the triangle is

(a) 6 cm (b) 8 cm
(c) 3 cm (d) 5 cm

[SSC, 2014]

52. A circular wire of diameter 112 cm is cut and bent in the form of a rectangle whose sides are in the ratio of 9 : 7. The small side of the rectangle is

(a) 67 cm (b) 87 cm
(c) 77 cm (d) 97 cm

[SSC, 2014]

53. A parallelogram has sides 60 m and 40 m and one of its diagonals is 80 m long. Its area is

(a) $400\sqrt{15}$ m^2 (b) $450\sqrt{15}$ m^2
(c) $500\sqrt{15}$ m^2 (d) $600\sqrt{15}$ m^2

[SSC, 2014]

54. A lawn is in the form of a rectangle having its breadth and length in the ratio 3:4. The area of the lawn is $\frac{1}{12}$ hectare. The breadth of the lawn is:

(a) 25 metres (b) 50 metres
(c) 75 metres (d) 100 metres

[SSC, 2013]

55. The area of a rectangle is thrice that of a square. The length of the rectangle is 20 cm and the breadth of the rectangle is $\frac{3}{2}$ times that of the side of the square. The side of the square (in cm) is:

(a) 10 (b) 20
(c) 30 (d) 60

[SSC, 2013]

56. The diagonals of a rhombus are 12 cm and 16 cm. The length of one side is:

(a) 8 cm (b) 6 cm
(c) 10 cm (d) 12 cm

[SSC, 2013]

57. The diameter of a circular wheel is 7 m. How many revolutions will it make in travelling 22 Km?

(a) 100 (b) 400
(c) 500 (d) 1000

[SSC, 2013]

58. The area of an equilateral triangle is $9\sqrt{3}$ m^2. The length (in m) of the median is:

(a) $2\sqrt{3}$ (b) $3\sqrt{3}$
(c) $3\sqrt{2}$ (d) $2\sqrt{2}$

[SSC, 2013]

59. How many tiles, each 4 decimetres square, will be required to cover the floor of a room 8 m long and 6 m broad?

(a) 200 (b) 260
(c) 280 (d) 300

[SSC, 2013]

60. The area of the circumcircle of an equilateral triangle is 3π cm^2. The perimeter of the triangle is:

(a) $3\sqrt{3}$ cm (b) 9 cm
(c) 18 cm (d) 3 cm

[SSC Assistant Grade III, 2013]

61. In ΔABC, $\angle A = 90°$ and $AD \perp BC$ where D lies on BC. If $BC = 8$ cm, $AC = 6$ cm, then $\Delta ABC : \Delta ACD$ is:

(a) 4:3 (b) 25:16
(c) 16:9 (d) 25:9

[SSC Assistant Grade III, 2013]

62. The sides of a triangle are in the ratio $\frac{1}{4}:\frac{1}{6}:\frac{1}{8}$ and its perimeter is 91 cm. The difference of the length of the longest side and that of the shortest side is:

(a) 19 (b) 20
(c) 28 (d) 21

[SSC Assistant Grade III, 2013]

63. The perimeter of an isosceles right-angled triangle is $2p$ cm. Its area is:

(a) $(3+2\sqrt{2})p$ cm^2 (b) $(3-2\sqrt{2})p^2$ cm^2
(c) $(2-\sqrt{2})p$ cm^2 (d) $(2+\sqrt{2})p^2$ cm^2

[SSC Assistant Grade III, 2013]

64. The ratio between the areas of two circles is 4:7. What will be the ratio of their radii?

(a) $2:\sqrt{7}$ (b) 4:7

(c) 16:49 (d) $4:\sqrt{7}$

[SSC Assistant Grade III, 2013]

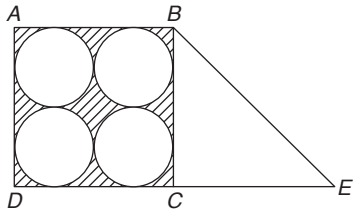
65. The perimeter of a non-square rhombus is 20 cm. One of its diagonals is 8 cm. The area of the rhombus is:

(a) 28 cm^2 (b) 20 cm^2

(c) 22 cm^2 (d) 24 cm^2

[SSC Assistant Grade III, 2013]

Directions (66–67): Study the following diagram to answer the questions.



66. If the diameter of each circle is 14 cm and $DC = CE$, the area of ΔBDE is:

(a) 784 cm^2 (b) 748 cm^2

(c) 874 cm^2 (d) 441 cm^2

(e) None of these

[IBPS PO/MT, 2013]

67. The area of the shaded region of square $ABCD$ is:

(a) 186 cm^2 (b) 168 cm^2

(c) 188 cm^2 (d) 441 cm^2

(e) None of these

[IBPS PO/MT, 2013]

68. What is the area of a given right-angled triangle?

I. The length of the hypotenuse is 5 cm.

II. The perimeter of the triangle is four times that of itsbase.

III. Oneof the angles of the triangle is 60°.

(a) Only II (b) Only III

(c) Either II or III (d) Both I and III

(e) Question cannot be answered even with the information in all three statements

[IBPS PO/MT, 2013]

69. The sides of a triangle are 50 cm, 78 cm and 112 cm. The smallest altitude is:

(a) 20 cm (b) 30 cm

(c) 40 cm (d) 50 cm

[SSC Assistant Grade III, 2012]

70. In a triangle ABC, $AB + BC = 12$ cm, $BC + CA = 14$ cm and $CA + AB = 18$ cm. Find the radius of the circle (in cm) which has the same perimeter as the triangle.

(a) $\frac{5}{2}$ (b) $\frac{7}{2}$

(c) $\frac{9}{2}$ (d) $\frac{11}{2}$

[SSC, 2012]

71. A playground is in the shape of a rectangle. A sum of ₹1,000 was spent to make the ground usable at the rate of 25 paisa per m^2. The breadth of the ground is 50 m. If the length of the ground is increased by 20 m, what will be the expenditure in rupees at the same rate per m^2?

(a) 1,250 (b) 1,000

(c) 1,500 (d) 2,250

[SSC, 2012]

72. The lengths of three medians of a triangle are 9 cm, 12 cm and 15 cm. The area (in cm^2) of the triangle is:

(a) 24 (b) 72

(c) 48 (d) 144

[SSC, 2012]

73. A circle and a rectangle have the same perimeter. The sides of the rectangle are 18 cm and 26 cm. The area of the circle is $\left[\text{Take } \pi = \frac{22}{7}\right]$

(a) 125 cm^2 (b) 230 cm^2

(c) 550 cm^2 (d) 616 cm^2

[SSC, 2012]

74. The area of a circle is increased by 22 cm^2 when its radius is increased by 1 cm. The original radius of the circle is:

(a) 3 cm (b) 5 cm

(c) 7 cm (d) 9 cm

[SSC, 2012]

75. The sum of all interior angles of a regular polygon is twice the sum of all its exterior angles. The number of sides of the polygon is:

(a) 10 (b) 8

(c) 12 (d) 6

[SSC, 2012]

76. If the diagonals of a rhombus are 8 and 6, then the square of its size is:

(a) 25 (b) 55

(c) 64 (d) 36

[SSC, 2012]

77. The area of the square inscribed in a circle of radius 8 cm is:

(a) 256 cm^2 (b) 250 cm^2
(c) 128 cm^2 (d) 125 cm^2

[SSC, 2012]

78. The area of a square is 1444 square metres. The breadth of a rectangle is $\frac{1}{4}$ the side of the square and the length of the rectangle is thrice its breadth. What is the difference between the area of the square and the area of the rectangle?

(a) 1152.38 m^2 (b) 1169.33 m^2
(c) 1181.21 m^2 (d) 1173.25 m^2
(e) None of these

[IBPS PO/MT, 2012]

79. What would be the cost of laying a carpet on a floor which has its length and breadth in the respective ratio of 32:21 and where its perimeter is 212 feet, if the cost per square foot of laying the carpet is ₹2.5?

(a) ₹6720 (b) ₹5420
(c) ₹7390 (d) Cannot be determined

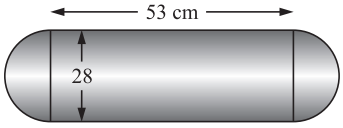
[Rajasthan Grameen Bank PO, 2011]

80. The circumference of 2 circles is 83 m and 220 m, respectively. What is the difference between the area of the larger circle and the smaller circle?

(a) 3422 m^2 (b) 3242 m^2
(c) 3244 m^2 (d) None of these

[Corporation Bank PO, 2011]

81. What is the area of the following figure?



(a) 2504 cm^2 (b) 1940 cm^2
(c) 2100 cm^2 (d) Cannot be determined

82. The length of a rectangle is twice the diameter of a circle. The circumference of the circle is equal to the area of a square of side 22 cm. What is the breadth of the rectangle if its perimeter is 668 cm?

(a) 24 cm (b) 26 cm
(c) 52 cm (d) Cannot be determined

[Union Bank of India PO, 2011]

83. A square is of area 200 m^2. A new square is formed in such a way that the length of its diagonal is $\sqrt{2}$ times of the diagonal of the given square. Then the area of the new square formed is:

(a) $200\sqrt{2}$ m^2 (b) $400\sqrt{2}$ m^2
(c) 400 m^2 (d) 800 m^2

[SSC, 2011]

84. If the length of a rectangular field is increased by 20% and the breadth is reduced by 20%, the area of the rectangle will be 192 m^2. What is the area of the original rectangle?

(a) 184 m^2 (b) 196 m^2
(c) 204 m^2 (d) 225 m^2
(e) None of these

[SBI Associates Banks PO, 2011]

85. Inside a square plot, a circular garden is developed which exactly fits in the square plot and the diameter of the garden is equal to the side of the square plot which is 28 m. What is the area of the space left out in the square plot after developing the garden?

(a) 98 m^2 (b) 146 m^2
(c) 84 m^2 (d) 168 m^2
(e) None of these

[SBI Associates Banks PO, 2011]

86. What is the cost of flooring a rectangular hall?

Statements:

I. The length of the rectangle is 6 meters.

II. The breadth of the rectangle is 2/3 of its length.

III. The cost of flooring the area of 100 cm^2 is ₹45.

(a) Only I and III
(b) Only II and III
(c) All I, II and III
(d) Question cannot be answered even with data in all three statements.
(e) None of these

[SBI Associates Banks PO, 2011]

87. The length of a rectangle is 3/5 of the side of a square. The radius of a circle is equal to side of the square. The circumference of the circle is 132 cm. What is the area of the rectangle if the breadth of the rectangle is 8 cm?

(a) 112.4 cm^2 (b) 104.2 cm^2
(c) 100.8 cm^2 (d) Cannot be determined
(e) None of these

[IOB PO, 2011]

88. The smallest side of a right-angled triangle is 8 cm less than the side of a square of perimeter 56 cm. The second largest side of the right-angled triangle is 4 cm less than the length of the rectangle of area 96 cm^2 and breadth 8 cm. What is the largest side of the right-angled triangle?

(a) 20 cm (b) 12 cm
(c) 10 cm (d) 15 cm
(e) None of these

[IOB PO, 2011]

89. The sum of the circumference of a circle and the perimeter of a square is equal to 272 cm. The diameter of the circle is 56 cm. What is the sum of the areas of the circle and the square?

(a) 2464 cm^2 (b) 2644 cm^2
(c) 3040 cm^2 (d) Cannot be determined
(e) None of these

[Allahabad Bank PO, 2011]

90. The largest and the second largest angles of a triangle are in the ratio of 4: 3. The smallest angle is half the largest angle. What is the difference between the smallest and the largest angles of the triangle?

(a) 30° (b) 60°
(c) 40° (d) 20°
(e) None of these

[Allahabad Bank PO, 2011]

91. The ratio of the 3 angles of a quadrilateral is 13:9:5. The value of the 4 angle of the quadrilateral is 36°. What is the difference between the largest and the second smallest angles of the quadrilateral?

(a) 104° (b) 108°
(c) 72° (d) 96°
(e) None of these

[Allahabad Bank PO, 2011]

92. The circumference of two circles is 88 metres and 220 metres respectively. What is the difference between the area of the larger circle and that of the smaller circle?

(a) 3422 m^2 (b) 3242 m^2
(c) 3244 m^2 (d) 3424 m^2
(e) None of these

[Corporation Bank PO, 2011]

93. The area of a square is 196 cm^2. whose side is half the radius of a circle. The circumference of the circle is equal to breadth of a rectangle, if perimeter of the rectangle is 712 cm. What is the length of the rectangle?

(a) 196 cm^2 (b) 186 cm^2
(c) 180 cm^2 (d) 190 cm^2

[OBC PO, 2010]

94. The sides of a right-angled triangle forming right angle are in the ratio 5:12. If the area of the triangle is 270 cm^2, then the length of the hypotenuse is:

(a) 39 cm (b) 42 cm
(c) 45 cm (d) 51 cm

[SSC, 2010]

95. If the measures of a diagonal and the area of a rectangle are 25 cm and 168 cm^2 respectively, what is the length of the rectangle?

(a) 31 cm (b) 24 cm
(c) 17 cm (d) 27 cm

[SSC, 2010]

96. A General, while arranging his men, who were 6000 in number, in the form of a square, found that there were 71 men leftover. How many were arranged in each row?

(a) 73 (b) 77
(c) 87 (d) 93

[SSC, 2010]

97. If the length of a rectangle is increased by 10% and its breadth is decreased by 10%, then its area:

(a) decreases by 1% (b) increases by 1%
(c) decreases by 2% (d) remains unchanged

[SSC, 2010]

98. If the length of a rectangle is increased in the ratio 6:7 and its breadth is diminished in the ratio 5:4 then its area will be diminished in the ratio:

(a) 17:16 (b) 15:14
(c) 9:8 (d) 8:7

[SSC, 2010]

99. 3 horses are tethered at 3 corners of a triangular plot of land having sides 20 m, 30 m and 40 m each with a rope of length 7 m. The area (in m^2) of the region of this plot, which can be grazed by the horses, is:

$\left(\text{Use } \pi = \frac{22}{7}\right)$

(a) $\frac{77}{3}$ (b) 75
(c) 77 (d) 80

[SSC, 2010]

100. A wire, when bent in the form of a square, encloses a region of area 121 cm^2. If the same wire is bent into the form of a circle, then the area of the circle is:

$\left(\text{Use } \pi = \frac{22}{7}\right)$

(a) 150 cm^2 (b) 152 cm^2
(c) 154 cm^2 (d) 159 cm^2

[SSC, 2010]

101. The ratio of the area of a sector of a circle to the area of the circle is 1:4. If the area of the circle is 154 cm^2, the perimeter of the sector is:

(a) 20 cm (b) 25 cm
(c) 36 cm (d) 40 cm

[SSC, 2010]

102. The sum of the areas of the 10 squares, the lengths of whose sides are 20 cm, 21 cm, ..., 29 cm respectively is:

(a) 6085 cm^2 (b) 8555 cm^2
(c) 2470 cm^2 (d) 11025 cm^2

[SSC, 2010]

103. The angles of a quadrilateral are in the ratio of 2:4:7:5. The smallest angle of the quadrilateral is equal to the smallest angle of a triangle. One of the angles of the triangle is twice the smallest angle of the triangle. What is the second largest angle of the triangle?

(a) 80° (b) 60°
(c) 120° (d) Cannot be determined
(e) None of these

[Central Bank of India PO, 2010]

104. The area of a square is 1024 cm^2. What is the ratio of the length to the breadth of a rectangle whose length is twice the side of the square and breadth is 12 cm less than the side of this square?

(a) 5:18 (b) 16:7
(c) 14:5 (d) 32:5
(e) None of these

[Central Bank of India PO, 2010]

105. The perimeter of a square is equal to twice the perimeter of a rectangle whose dimensions are: length 8 cm and breadth 7 cm. What is the circumference of a semicircle whose diameter is equal to the side of the square?

(Rounded off of the decimal place)

(a) 38.57 cm (b) 23.57 cm
(c) 42.46 cm (d) 47.47 cm
(e) None of these

[Punjab and Sind Bank PO, 2010]

106. The circumferences of two circles are 132 metres and 176 metres respectively. What is the difference between the area of the larger circle and that of the smaller circle?

(a) 1048 m^2 (b) 1076 m^2
(c) 1078 m^2 (d) 1090 m^2
(e) None of these

[Indian Bank PO, 2010]

ANSWER KEYS

EXERCISE–1

1. (c)	**2.** (a)	**3.** (b)	**4.** (b)	**5.** (c)	**6.** (a)	**7.** (b)	**8.** (c)	**9.** (a)	**10.** (c)	**11.** (c)	**12.** (c)	**13.** (c)
14. (c)	**15.** (a)	**16.** (c)	**17.** (c)	**18.** (b)	**19.** (a)	**20.** (b)	**21.** (b)	**22.** (c)	**23.** (b)	**24.** (c)	**25.** (c)	**26.** (a)
27. (c)	**28.** (a)	**29.** (c)	**30.** (b)	**31.** (a)	**32.** (b)	**33.** (c)	**34.** (a)	**35.** (b)	**36.** (a)	**37.** (a)	**38.** (c)	**39.** (b)
40. (b)	**41.** (a)	**42.** (d)	**43.** (d)	**44.** (c)	**45.** (a)	**46.** (c)	**47.** (d)	**48.** (c)	**49.** (c)	**50.** (d)	**51.** (c)	**52.** (b)
53. (c)	**54.** (c)	**55.** (a)	**56.** (a)	**57.** (c)	**58.** (c)	**59.** (b)	**60.** (b)	**61.** (a)	**62.** (c)	**63.** (b)	**64.** (a)	**65.** (d)
66. (d)	**67.** (b)	**68.** (c)	**69.** (c)	**70.** (c)	**71.** (c)	**72.** (b)	**73.** (c)	**74.** (b)	**75.** (b)	**76.** (d)	**77.** (b)	**78.** (c)
79. (d)	**80.** (a)	**81.** (b)	**82.** (d)	**83.** (a)	**84.** (b)	**85.** (b)	**86.** (d)	**87.** (a)	**88.** (c)	**89.** (a)	**90.** (c)	**91.** (b)
92. (a)	**93.** (b)	**94.** (c)	**95.** (b)	**96.** (a)	**97.** (b)	**98.** (b)	**99.** (b)	**100.** (d)	**101.** (c)	**102.** (b)	**103.** (d)	**104.** (c)
105. (c)	**106.** (b)											

EXERCISE–2

1. (c)	**2.** (b)	**3.** (b)	**4.** (a)	**5.** (c)	**6.** (c)	**7.** (c)	**8.** (a)	**9.** (d)	**10.** (b)	**11.** (d)	**12.** (c)	**13.** (a)
14. (d)	**15.** (c)	**16.** (b)	**17.** (c)	**18.** (d)	**19.** (a)	**20.** (a)	**21.** (d)	**22.** (b)	**23.** (a)	**24.** (c)	**25.** (c)	**26.** (d)
27. (d)	**28.** (b)	**29.** (d)	**30.** (b)	**31.** (a)	**32.** (d)	**33.** (c)	**34.** (d)	**35.** (a)	**36.** (b)	**37.** (a)	**38.** (a)	**39.** (b)
40. (c)	**41.** (a)	**42.** (d)	**43.** (d)	**44.** (a)	**45.** (c)	**46.** (d)	**47.** (a)	**48.** (b)	**49.** (b)	**50.** (c)	**51.** (d)	**52.** (c)
53. (d)	**54.** (a)	**55.** (a)	**56.** (c)	**57.** (d)	**58.** (b)	**59.** (d)	**60.** (b)	**61.** (c)	**62.** (d)	**63.** (b)	**64.** (a)	**65.** (d)
66. (a)	**67.** (b)	**68.** (d)	**69.** (b)	**70.** (b)	**71.** (a)	**72.** (b)	**73.** (d)	**74.** (a)	**75.** (d)	**76.** (a)	**77.** (c)	**78.** (d)
79. (a)	**80.** (d)	**81.** (c)	**82.** (b)	**83.** (c)	**84.** (e)	**85.** (d)	**86.** (c)	**87.** (c)	**88.** (c)	**89.** (c)	**90.** (c)	**91.** (d)
92. (e)	**93.** (c)	**94.** (a)	**95.** (b)	**96.** (b)	**97.** (a)	**98.** (b)	**99.** (c)	**100.** (c)	**101.** (b)	**102.** (a)	**103.** (b)	**104.** (e)
105. (a)	**106.** (c)											

EXPLANATORY ANSWERS

EXERCISE-1

1. (c) Let, a = 15 m, b = 16 m, c = 17 m.

Then, $s = \frac{a+b+c}{2} = \frac{15+16+17}{2} = 24$ m.

$\therefore$ Area $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{24(24-15)(24-16)(24-17)}$

$= \sqrt{24\times 9\times 8\times 7}$

$= \sqrt{12096} = 24\sqrt{21}$ m^2.

2. (a) Height of the triangle $= \sqrt{(6.5)^2 - 6^2}$

$= \sqrt{42.25-36}$

$= \sqrt{6.25} = 2.5$ m

$\therefore$ Area $= \frac{1}{2}$ (base $\times$ height)

$= \frac{1}{2} \times 6 \times 2.5 = 7.5$ m^2.

3. **(b)** Each side = 12 cm

Then, $s = \frac{12+12+12}{2} = 18$.

Area $= \sqrt{18(18-12)(18-12)(18-12)}$

$= \sqrt{18\times6\times6\times6} = \frac{1}{2} \times 12 \times$ height

or, height $= \frac{36\sqrt{3}}{6} = 6\sqrt{3}$ cm.

4. **(b)** Let, the length of equal sides be x.

Then, $s = \frac{x+x+64}{2} = x + 32$.

Area = 1600 m^2.

$= \sqrt{(x+32)(x+32-x)(x+32-x)(x+32-64)}$

$= \sqrt{(x+32)\times32\times32\times(x-32)}$

or, $1600 = 32\sqrt{x^2-32^2}$

$\Rightarrow \sqrt{x^2-32^2} = 50$

or, $x^2 = 32^2 + 50^2 = 1024 + 2500 = 3524$.

$\therefore$ $x = 59.36$ m.

5. **(c)** Here, $s = \frac{20+21+29}{2} = \frac{70}{2} = 35$ m.

$\therefore$ Area $= \sqrt{35(35-20)(25-21)(35-29)}$

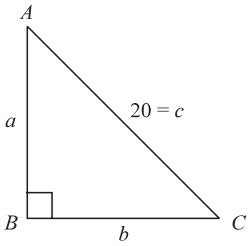
$= \sqrt{35\times15\times14\times6}$

$= \sqrt{5^2\times7^2\times3^2\times2^2}$

$= 5 \times 7 \times 3 \times 2 = 210$ m^2.

6. **(a)** We have, $a + b + 20 = 2 \times 24$

$\therefore$ $a + b = 28$ …(1)



By Pythagores Theorem, $a^2 + b^2 = 400$

$\because$ $(a + b)^2 = 28^2$

$\Rightarrow ab = \frac{28^2-(a^2+b^2)}{2} = \frac{28^2-400}{2} = 192$

$\therefore$ $a - b = \sqrt{(a+b)^2-4ab}$

$= \sqrt{784-4\times192} = 4$ …(2)

Solving (1) and (2), we get

$a = 16$ cm and $b = 12$ cm.

7. **(b)** The sides of the triangle are

$a = \frac{3}{12} \times 36 = 9$ cm, $b = \frac{4}{12} \times 36 = 12$ cm

$c = \frac{5}{12} \times 36 = 15$ cm

$s = \frac{a+b+c}{2} = \frac{36}{2} = 18$ cm

$\therefore$ Area $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{18(18-9)(18-12)(18-15)}$

$= \sqrt{18\times9\times6\times3}$

$= \sqrt{9^2\times2^2\times3^2} = 9 \times 2 \times 3 = 54$ cm^2.

8. **(c)** We have, $2s = 50 + 78 + 112 = 240$

$\therefore$ $s = 120$.

$\therefore$ Area $= \sqrt{120(120-50)(120-78)(120-112)}$

$= \sqrt{120\times70\times42\times8}$

$= \sqrt{2^3\times5\times3\times7\times5\times2\times2\times7\times3\times2^3}$

$= 2^4 \times 5 \times 3 \times 7 = 1680$ m^2

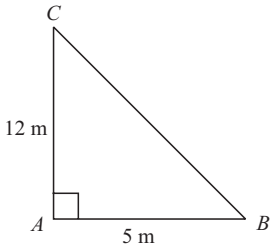
$\therefore$ $\frac{1}{2} \times 112 \times h = 1680$,

where h is the length of perpendicular

$\therefore$ $h = \frac{1680\times2}{112} = 30$ m.

9. **(a)** Let, BC be the ladder.

Then, $BC = \sqrt{12^2+5^2} = 13$ m.

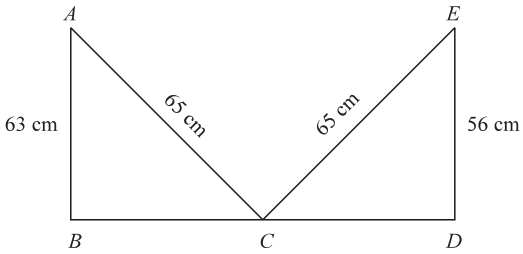


10. **(c)** $BC = \sqrt{(65)^2-(63)^2} = 16$

$CD = \sqrt{(65)^2-(56)^2} = 33$.

$\therefore$ Width of the street

$= 16 + 33 = 49$ cm.



11. **(c)** Given: $\frac{1}{2} \times x \times h = x^2$

$\therefore$ $h = 2x$.

12. (c) Let, the base be $3x$.

Then, the height is $4x$.

Given: $\frac{1}{2} \times 3x \times 4x = 150 \Rightarrow x = 5$.

$\therefore$ Base = $3 \times 5 = 15$ m and height = $4 \times 5 = 20$ m.

13. (c) Area of field = $\frac{495.72}{36.72} \times 10000$

$= 135000 \text{ m}^2$

Let, the height be x.

Then, base = $3x$.

We have, $\frac{1}{2} \times x \times 3x = 135000$

$\Rightarrow \quad x^2 = 90000$ or $x = 300$.

$\therefore$ Height = 300 m and base = $3 \times 300 = 900$ m.

14. (c) Let, the original sides be a, b, c, then

$s = \frac{1}{2}(a + b + c)$

and area of the triangle = $\sqrt{s(s-a)(s-b)(s-c)}$

For the new triangle, the sides are $2a$, $2b$, $2c$.

Then, $S = \frac{1}{2}(2a + 2b + 2c) = a + b + c = 2s$.

$\therefore$ Area of new triangle

$= \sqrt{S(S-2a)(S-2b)(S-2c)}$

$= \sqrt{2s(2s-2a)(2s-2b)(2s-2c)}$

$= \sqrt{16s(s-a)(s-b)(s-c)}$

$= 4\sqrt{s(s-a)(s-b)(s-c)}$

$= 4 \times$ (area of original triangle).

15. (a) The third side of the triangle

$= 324 - (85 + 154) = 85$ m.

Also, $s = \frac{a+b+c}{2} = \frac{324}{2} = 162$.

$\therefore$ Area of the triangle

$= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{162(162-85)(162-85)(162-154)}$

$= \sqrt{162 \times 77 \times 77 \times 8} = 2772 \text{ m}^2$.

$\therefore$ The cost of ploughing the field

$= 2772 \times 10 = ₹27720$.

16. (c) The area of the equilateral triangle

$= \frac{\sqrt{3}}{4}(\text{side})^2 = \frac{\sqrt{3}}{4} \times (2\sqrt{3})^2$

$= 3\sqrt{3} \text{ cm}^2$.

17. (a) Height of the equilateral triangle = $\frac{\sqrt{3}}{2} \times$ (side)

$\Rightarrow \quad 2\sqrt{3} = \frac{\sqrt{3}}{2} \times$ (side)

$\therefore$ Side of the equilateral triangle = 4 cm.

18. (b) $3 \times$ (side) $= 12 \Rightarrow$ side = 4 m.

$\therefore$ Area of equilateral triangle = $\frac{\sqrt{3}}{4}(\text{side})^2$

$= \frac{\sqrt{3}}{4} \times 16 = 4\sqrt{3}$ m.

19. (a) $3 \times$ (side) $= 24 \Rightarrow$ side = 8 cm.

$\therefore$ Height of the equilateral triangle

$= \frac{\sqrt{3}}{2} \times (\text{side}) = \frac{\sqrt{3}}{2} \times 8 = 4\sqrt{3}$ cm.

20. (b) Let, x and $(51 - x)$ be the other two sides of the triangle

Then, $x^2 + (51 - x)^2 = 39^2$

$\Rightarrow \quad x^2 + 2601 - 102x + x^2 = 1521$

$\Rightarrow \quad x = \frac{51 \pm \sqrt{441}}{2} = \frac{51 \pm 21}{2} = 36, 15.$

$\therefore$ The other two sides are 36 cm and 15 cm.

21. (b) Given: $a = \frac{5}{8}b$.

Now, perimeter of isosceles triangle = $2a + b$

$\Rightarrow 306 = 2 \times \frac{5}{8}b + b$ or, $b = 136$.

$\therefore \quad a = \frac{5}{8}b = \frac{5}{8} \times 136 = 85.$

$\therefore$ Area of isosceles triangle = $\frac{b}{4}\sqrt{4a^2 - b^2}$

$= \frac{136}{4}\sqrt{4 \times (85)^2 - (136)^2}$

$= 34 \times 102 = 3468 \text{ m}^2$.

22. (c) We have, hypotenuse = $\sqrt{2}\,a = 8 \Rightarrow a = 8/\sqrt{2}$.

$\therefore$ Area of isosceles triangle = $\frac{1}{2}a^2$

$= \frac{1}{2} \times \frac{64}{2} = 16 \text{ cm}^2$.

23. (b) Let, the lateral side = $5x$ and the base = $4x$.

Then, $5x + 5x + 4x = 14 \Rightarrow x = 1$.

$\therefore$ The sides of the triangle are 5 cm, 5 cm and 4 cm.

$\therefore$ Area of the isosceles triangle

$= \frac{b}{4}\sqrt{4a^2 - b^2}$

$= \frac{4}{4}\sqrt{4(5)^2 - (4)^2} = \sqrt{84} = 2\sqrt{21}$ cm^2.

24. (c) Area $= \sqrt{s(s-a)(s-b)(s-c)} = A$

where $s = \frac{a+b+c}{2}$ and a, b, c are sides of the triangle.

When the sides are increased by 200%, the sides become $3a$, $3b$ and $3c$.

$s_1 = \frac{3a+3b+3c}{2} = 3\frac{(a+b+c)}{2} = 3s.$

$A_1 = \sqrt{s_1(s_1-3a)(s_1-3b)(s_1-3c)}$

$= \sqrt{3s.3(s-a).3(s-b).3(s-c)}$

$= 9\sqrt{s(s-a)(s-b)(s-c)} = 9A.$

$\therefore$ Increase in area $= 9A - A = 8A$ or, 800%

25. (c) Let, the length of each of equal sides of the triangle be x m.

Then, $x^2 + x^2 = (50\sqrt{2})^2 = 5000.$

$\Rightarrow 2x^2 = 5000 \Rightarrow x = 50.$

$\therefore$ Perimeter of the triangle $= 50 + 50 + 50\sqrt{2}$

$= 100 + 50 \times 1.4146 = 170.73$ m.

$\therefore$ Cost of fencing $= ₹(170.73 \times 3) = ₹512.19.$

26. (a) Area of isosceles triangle $= \frac{b}{4}\sqrt{4a^2 - b^2}$

$= \frac{16}{4}\sqrt{4\times(10)^2 - (16)^2} = 4 \times 12 = 48$ cm^2

Given: Area of equilateral triangle = 48.

$\Rightarrow \frac{\sqrt{3}}{4}(\text{side})^2 = 48$

$\Rightarrow$ Side of equilateral triangle = 10.5 cm.

27. (c) Perimeter of a right-angled isosceles triangle

$= (\sqrt{2}+1) \times$ hypotenuse

$\Rightarrow 4\sqrt{2} + 4 = (\sqrt{2} + 1) \times$ hypotenuse

$\Rightarrow$ hypotenuse = 4 m.

28. (a) Here, $a = 60$, $b = 40$ and $d = 80$

$\therefore s = \frac{a+b+d}{2} = \frac{60+40+80}{2} = 90.$

$\therefore$ Area of the parallelogram

$= 2\sqrt{s(s-a)(s-b)(s-d)}$

$= 2\sqrt{90(90-60)(90-40)(90-80)}$

$= 2\sqrt{90\times30\times50\times10}$

$= 600\sqrt{15}$ m^2.

29. (c) Area of the parallelogram

$= 16 \times 14 = 224$ cm^2.

30. (b) We have, $s = \frac{a+b+d}{2}$

$= \frac{65+119+156}{2} = 170.$

$\therefore$ Area of parallelogram

$= 2\sqrt{s(s-a)(s-b)(s-d)}$

$= 2\sqrt{170(170-65)(170-119)(170-156)}$

$= 2\sqrt{170\times51\times105\times14}$

$= 2 \times 3570 = 7140$ m^2.

$\therefore$ Cost of gravelling $= 7140 \times 10 = ₹71400.$

31. (a) The area of the parallelogram

$= 10 \times 7 = 70$ m^2.

32. (b) Area of the parallelogram $= 8 \times 4 = 32$ m^2.

Distance between the shorter sides $= \frac{32}{5} = 6.4$ m.

33. (c) Area of quadrilateral $= \frac{1}{2}d(p_1 + p_2)$

$\Rightarrow 420 = \frac{1}{2}\times d \times (18+12)$

$\Rightarrow d = \frac{420\times2}{30} = 28$ m.

34. (a) Let, the base be x and altitude be $2x$.

Then, $x \times 2x = 72 \Rightarrow x^2 = 36 \Rightarrow x = 6.$

35. (b) Let, the base be x.

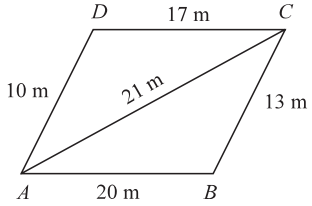
Area of the parallelogram = base $\times$ altitude

$\Rightarrow 240 = x \times 12$

$\therefore x = \frac{240}{12} = 20$ cm.

36. (a) Area of quadrilateral $ABCD$

= Area of ΔADC + Area of ΔABC, where,



Area of ΔABC

$= \sqrt{s(s-AB)(s-BC)(s-d)}$

$\left(s = \frac{AB+BC+d}{2} = \frac{20+13+21}{2} = 27\right)$

$= \sqrt{27(27-20)(27-13)(27-21)}$

$= \sqrt{27\times7\times14\times6} = 126 \text{ m}^2.$

Area of ΔADC

$= \sqrt{s(s-AD)(s-DC)(s-d)}$

$\left(s = \frac{AD+DC+d}{2} = \frac{10+17+21}{2} = 24\right)$

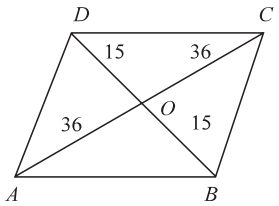
$= \sqrt{24(24-10)(24-17)(24-21)}$

$= \sqrt{24\times14\times7\times3} = 84 \text{ m}^2.$

Here, the area of quadrilateral $ABCD$

$= 126 + 84 = 210 \text{ m}^2.$

37. (a) Since $AC = 72$ and $BD = 30$



$\therefore \quad OB = OD = \frac{BD}{2} = \frac{30}{2} = 15$

and, $OA = OC = \frac{AC}{2} = \frac{72}{2} = 36$

$\therefore \quad AB = BC = CD = DA = \sqrt{36^2+15^2}$

$= \sqrt{1296+225}$

$= \sqrt{1521} = 39.$

$\therefore$ Perimeter of the parallelogram $= 4 \times 39$

$= 156$ cm.

38. (c) Area of the parallelogram

= base × altitude

$\Rightarrow \quad (x^2 - 4) = (x + 4) \times (x - 3)$

$\Rightarrow \quad x^2 - 4 = x^2 + 4x - 3x - 12$

$\Rightarrow \quad x = 8.$

$\therefore$ Area of the parallelogram

$= x^2 - 4 = (8)^2 - 4 = 60$ sq units.

39. (b) We have,

$d_1^2 + d_2^2 = 2(a^2 + b^2)$

$\Rightarrow \quad (16)^2 + d_2^2 = 2(12^2 + 14^2)$

$\Rightarrow \quad d_2^2 = 2(144 + 196) - 256 = 424.$

$\therefore \quad d^2_2 = \sqrt{424} = 20.6$ cm.

40. (b) $\pi r^2 = 154 \text{ m}^2$

$\Rightarrow r^2 = \frac{7}{22} \times 154 \text{ m}^2 = 49 \text{ m}^2$

$\therefore \quad r = 7$ m

Perimeter $= 2 \times \frac{22}{7} \times 7$ m $= 44$ m.

41. (a) If x be the side of the square and r be the radius of the circle, then,

$4x = 2\pi r$

or, $\quad x = \frac{\pi r}{2}$

Now, $\pi r^2 :: x^2 = \pi r^2 : : \frac{\pi^2 r^2}{4}$ or, $4:\pi$

$= 4 : : \frac{22}{7}$ or, 14:11.

42. (d) Let, the width of the rectangle be x m

Then length $= 3x$ m

Perimeter $= 2(x + 3x) = 96$

$\Rightarrow \quad 8x = 96$ or, $x = 12$

Area $= 12 \times 36 = 432 \text{ m}^2.$

43. (d) As the cow is tied at the corner of a rectangular field, it will graze the area of the field enclosed between the two sides of the rectangle.

$= \frac{1}{4}(\pi \times 14 \times 14)$

$= \frac{1}{4} \times \frac{22}{7} \times 14 \times 14$

$= 154 \text{ m}^2.$

44. (c) The distance covered by the wheel in one minute

$= \frac{66 \times 1000 \times 100}{60} = 110000$ cm

The distance covered by the wheel in one revolution

= The circumference of the wheel

$= 2\pi r = 2 \times \frac{22}{7} \times \frac{70}{2} = 220$ cm

$\therefore$ Number of revolutions of the wheel

$= \frac{110000}{220} = 500.$

45. (a) $\pi(8)^2 + \pi(6)^2 = \pi r^2$

$\therefore \quad r^2 = 64 + 36$

$r = 10$

46. (c) Area of two paths $= 2 \times (65 + 50 - 2) = 226 \text{ m}^2$

Cost of construction $= 226 \times 17.25$

$= ₹3898.50.$

47. (d) $\frac{1}{2}(75 + x) \times 40 = 2500$

$\Rightarrow \quad 75 + x = 125$

$\Rightarrow \quad x = 50$

$\therefore$ The other parallel side $= 50$ m.

48. (c) Let, the breadth be x, then length $= x + 3$

Given: $2(x + x + 3) = x(x + 3)$

$\Rightarrow \quad 4x + 6 = x^2 + 3x$

$\Rightarrow \quad 4x + 6 = x^2 + 3x$

$\Rightarrow \quad x^2 - x - 6 = 0$

$\therefore \quad x = \dfrac{1 \pm \sqrt{1+24}}{2} = \dfrac{1 \pm 5}{2}$

$\therefore \quad x = 3, -2$

$\therefore$ Breadth = 3 cm.

49. (c) Let, x, y be the sides of squares

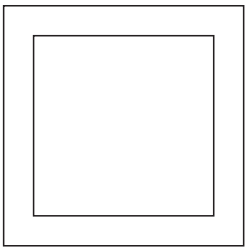
$\dfrac{x^2}{y^2} = \dfrac{9}{1} \Rightarrow \dfrac{x}{y} = \dfrac{3}{1}$

$\therefore$ The ratio of perimeters is $4x:4y$

i.e., $x:y = 3:1$.

50. (d) Let, x (in metres) be the width of the path

Side of outer square = $30 + 2x$



$\therefore$ Area of path = $(30 + 2x)^2 - 30^2$

$\therefore \quad (30 + 2x)^2 - 30^2 = 256$

$\Rightarrow \quad 4x^2 + 120x - 256 = 0$

$\Rightarrow \quad x^2 + 30x - 64 = 0$

$\Rightarrow \quad (x - 2)(x + 32) = 0$

$\therefore \quad x = 2 \qquad (\because x < 0)$

51. (c) Additional grassy ground grazed

$= \pi(23^2 - 12^2)$ m^2

$= \dfrac{22}{7} \times 35 \times 11$

$= 1210$ m^2.

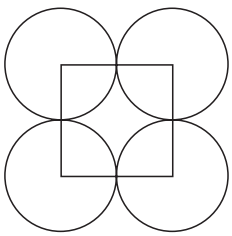
52. (b) Required area enclosed

$= 14 \times 14 - 4 \times$ (area of a quadrant)

$= \left[196 - 4 \times \dfrac{22}{7} \times 7 \times 7 \times \dfrac{90}{360}\right]$ cm^2

$= (196 - 154)$ cm^2

$= 42$ cm^2.



53. (c) Area = $\dfrac{6000 \times 100}{25} = 24000$ m^2

Let, the length be $5x$ and breadth be $3x$

$\therefore \quad 5x \times 3x = 24000$

$\therefore \quad x = \sqrt{1600} = 40$

$\therefore$ Length = $5 \times 40 = 200$ m, breadth = 3×40

$= 120$ m.

54. (c) Area of the carpet = $\dfrac{105}{3.50} = 30$ m^2

Area of the room = 30 m^2

Width = 5 m

$\therefore$ Length = $\dfrac{30}{5} = 6$ m.

55. (a) Area of the rectangular field

$= \dfrac{28000}{3500} = 8$ hectares

$2x \times x = 80000$

$\therefore \quad x = \sqrt{40000} = 200$

$\therefore$ Breadth = 200 m

Length = 400 m

Perimeter = $2(400 + 200) = 1200$ m

$\therefore$ Cost of fencing = $1200 \times 5 = ₹6000$.

56. (a) Let, the length and breadth be $6x$ and $5x$ m respectively. Then

$6x \times 5x = 27000$ or $30x^2 = 27000$

$\Rightarrow \quad x^2 = 900$ or $x = 30$

Hence, length of the field = 180 m and width = 150 m.

57. (c) Area of sector = $\left(\dfrac{1}{2} \times \text{arc length} \times \text{radius}\right)$ cm^2

$= \left(\dfrac{1}{2} \times 3.5 \times 5\right)$ cm^2

$= 8.75$ cm^2.

58. (c) Area of the plot = $150 \times 3 = 450$ m^2

If the length of the plot be x m then breadth

$= \dfrac{x}{2}$ m

$\therefore \quad x \times \dfrac{x}{2} = 450$

or, $x^2 = 900$

or, $x = 30$ m.

59. (b) Perimeter of circle = $2 \times \dfrac{22}{7} \times 7$

$= 44$ cm

Perimeter of half circle = 22 cm

The length of the wire = $22 + 14 = 36$ cm.

60. **(b)** $2\pi R - 2\pi r = 66$

$2\pi(R - r) = 66$

$2 \times \frac{22}{7} \times (R - r) = 66$

or, $R - r = 66 \times \frac{7}{22} \times \frac{1}{2} = \frac{21}{2} = 10.5$ m.

61. **(a)** One side of the square $= \sqrt{50} = 5\sqrt{2}$

Length of diagonal $= 5\sqrt{2} \times \sqrt{2} = 10$

Radius of circle $= 5$

$\therefore$ Area of the circle $= 25\pi$ units.

62. **(c)** Area of remaining portion

= Area of rectangle − Area of circle

$= 4 \times 2 - \pi \times (1)^2 = (8 - \pi)$ cm^2.

63. **(b)** $r = \sqrt{22^2 + 19^2 + 8^2}$ cm

$= \sqrt{484 + 361 + 64}$ cm

$= \sqrt{909}$ cm $= 30$ cm

64. **(a)** Area of rectangle $= l \times b$

Let, the new width be b_1

Then, $l \times b = \frac{4}{3} l \times b_1$

$\therefore \quad b_1 = \frac{3}{4} b = 0.75b$

Thus, there should be a reduction of 25% in the width.

65. **(d)** Area of rectangle $= 6.4 \times 2.5 = 16$ m^2

According to question:

Area of square = Area of the rectangle

$\therefore$ Area of square $= 16$ m^2

$\therefore$ Side of the square $= 4$ m.

66. **(d)** $\pi \times 5 \times 5 - \pi \times 4 \times 4$

$= 25\pi - 16\pi = 9\pi$ cm^2.

67. **(b)** The length of the room $= 4$ m

Since it can be partitioned into two equal square rooms, the two equal parts can only be of 2 m each.

68. **(c)** Let length $= a$ m and breadth $= b$ m

Then, $2(a + b) = 46$

or, $a + b = 23$ and $ab = 120$

$\therefore$ Diagonal $= \sqrt{a^2 + b^2} = \sqrt{(a+b)^2 - 2ab}$

$= \sqrt{(23)^2 - 2 \times 120} = \sqrt{289} = 17$ m.

69. **(c)** Perimeter of the square $= 4 \times 22$ m $= 88$ m

$2\pi r = 88$ m

$\Rightarrow \quad r = 88 \times \frac{7}{22 \times 2}$ m $= 14$ m.

70. **(c)** Area of an equilateral $\Delta = \frac{\sqrt{3}}{4} a^2$

$[3a = 132 \ \therefore \ a = 44]$

$= \frac{\sqrt{3}}{4} \times 44 \times 44$

$= 838.312$ m^2.

Area of square $= a^2 = 33 \times 33$

$= 1089$ m^2 $\left[a = \frac{132}{4} = 33\right]$

Area of regular hexagon

$= \frac{3\sqrt{3}a^2}{2} = \frac{3\sqrt{3} \times 22 \times 22}{2}$. $\left[a = \frac{132}{6} = 22\right] = 1257.47$

Area of circle $= \pi r^2$

$= \frac{22}{7} \times 21 \times 21$ $\left[r = \frac{132}{2\pi} = 21\right] = 1386$

$\therefore$ Circle has largest surface area.

71. **(c)** Let, r be the radius of smaller circle

Radius of larger circle $= 12r$

$\frac{\text{Area of larger circle}}{\text{Area of smaller circle}} = \frac{\pi(12r)^2}{\pi r^2} = \frac{144}{1}$

$\therefore$ Area of large circle contains the area of smaller circle 144 times.

72. **(b)** If a be the side of the square and r be the radius of the circle, then,

$2\pi r = 4a$

or, $r = \frac{4a}{2\pi} = \frac{2a}{\pi}$

$\therefore$ Area of the circle $= \pi r^2$ and,

area of the square $= a^2$

$\therefore$ Area of the circle:Area of the square

$= \pi r^2 : a^2$

$= \frac{\pi\left(\frac{2a}{\pi}\right)^2}{a^2} = \frac{\pi \frac{4a^2}{\pi^2}}{a^2}$

$= \frac{4}{\pi} = \frac{4 \times 7}{22} = 14:11.$

73. **(c)** Let, the length and breadth of the plot be $3x$ and x feet, respectively.

Total area of the plot $= 4 \times 1200 = 4800$ ft^2.

$\therefore \quad x \times 3x = 4800 \Rightarrow x = 40$ ft

$\therefore$ Length $= 3 \times 40 = 120$ ft.

74. **(b)** Ratio of the areas = 4:25

Ratio of the sides = 2:5

If the side of s_1 is 2 cm, then side of s_2 is 5 cm.

If the side of s_1 is 6 cm, the side of s_2 is $\frac{5}{2} \times 6 = 15$ cm.